Version 7.3.1

PVsyst - Simulation report

Grid-Connected System

Project: kolkatawith high rise

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 7.04 kWp

Taliganja - India

Author

Project summary

Geographical Site
Taliganja
India

Situation

Latitude	22.48 °N
Longitude	88.35 °E
Altitude	10 m
Time zone	UTC+5.5

Project settings
Albedo 0.20

Meteo data
Taliganja
Meteonorm 8.1 (1991-2012) - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation
Fixed plane
Tilt/Azimuth 21.9 / 0 °

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	22 units
Pnom total	7.04 kWp

Inverters

Nb. of units	2 units
Pnom total	5.00 kWac
Pnom ratio	1.408

Results summary

Produced Energy	9583 kWh/year	Specific production	1361 kWh/kWp/year	Perf. Ratio PR	82.13 %

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General parameters

Grid-Connected System | **No 3D scene defined, no shadings**

PV Field Orientation

Orientation		**Sheds configuration**	**Models used**	
Fixed plane		No 3D scene defined	Transposition	Perez
Tilt/Azimuth	21.9 / 0 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate
Horizon		**Near Shadings**	**User's needs**	
Free Horizon		No Shadings	Unlimited load (grid)	

PV Array Characteristics

PV module		**Inverter**	
Manufacturer	GCL	Manufacturer	Delta Energy
Model	GCL-P6/72-320	Model	Solar Inverter RPI H2.5 Flex
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	320 Wp	Unit Nom. Power	2.50 kWac
Number of PV modules	22 units	Number of inverters	2 units
Nominal (STC)	7.04 kWp	Total power	5.0 kWac
Modules	2 Strings x 11 In series	Operating voltage	125-550 V
At operating cond. (50°C)		Max. power (=>25°C)	3.00 kWac
Pmpp	6.37 kWp	Pnom ratio (DC:AC)	1.41
U mpp	369 V		
I mpp	17 A		
Total PV power		**Total inverter power**	
Nominal (STC)	7 kWp	Total power	5 kWac
Total	22 modules	Number of inverters	2 units
Module area	43.6 m²	Pnom ratio	1.41
Cell area	39.9 m²		

Array losses

Thermal Loss factor		**DC wiring losses**		**Module Quality Loss**	
Module temperature according to irradiance		Global array res.	357 mΩ	Loss Fraction	-0.8 %
Uc (const)	20.0 W/m²K	Loss Fraction	1.5 % at STC		
Uv (wind)	0.0 W/m²K/m/s				
Module mismatch losses		**Strings Mismatch loss**		**IAM loss factor**	
Loss Fraction	2.0 % at MPP	Loss Fraction	0.1 %	ASHRAE Param.: IAM = 1 - bo (1/cosi -1)	
				bo Param.	0.05

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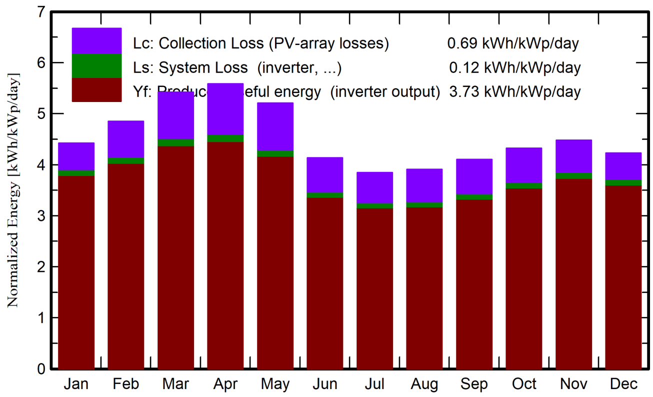
Main results

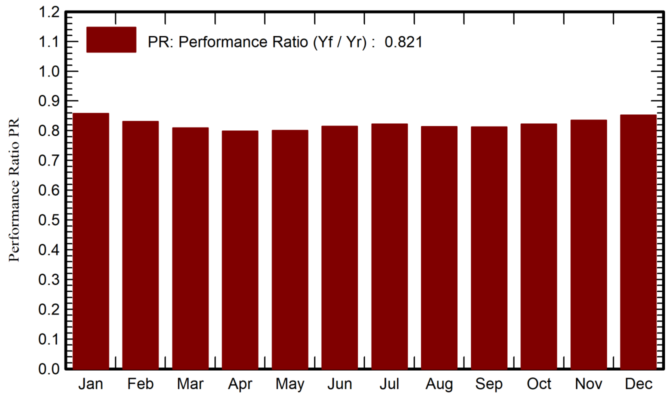
System Production

Produced Energy	9583 kWh/year	Specific production	1361 kWh/kWp/year
		Performance Ratio PR	82.13 %

Normalized productions (per installed kWp)



Performance Ratio PR



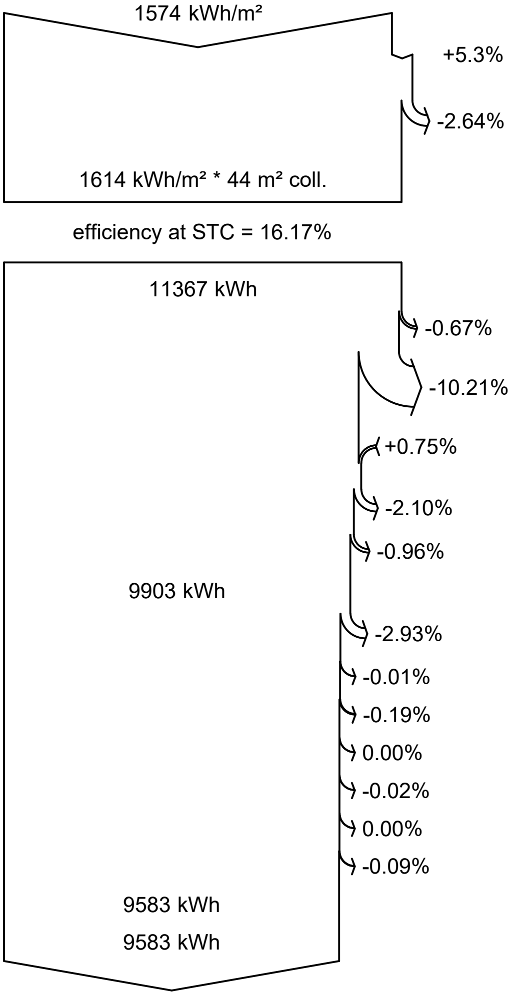
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	112.6	61.13	17.32	137.2	133.8	853.0	827.9	0.857
February	118.4	63.40	21.96	135.9	132.9	819.7	795.1	0.831
March	157.1	82.81	27.48	168.0	164.1	986.1	956.4	0.808
April	167.1	90.50	30.21	167.6	163.5	970.9	941.7	0.798
May	171.3	99.30	31.52	161.5	157.1	938.8	910.3	0.801
June	133.8	94.20	30.46	124.1	120.1	734.0	711.2	0.814
July	128.1	95.88	29.70	119.3	115.1	712.1	690.1	0.822
August	124.9	84.40	29.26	121.1	117.5	716.4	693.7	0.814
September	120.3	72.48	28.19	123.1	119.8	726.6	703.9	0.812
October	122.0	76.68	27.12	134.0	130.5	798.9	775.1	0.822
November	112.7	60.79	23.21	134.4	131.2	814.7	790.1	0.835
December	106.2	57.01	18.54	131.1	128.1	811.9	787.1	0.853
Year	1574.4	938.59	26.26	1657.3	1613.5	9883.0	9582.6	0.821

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

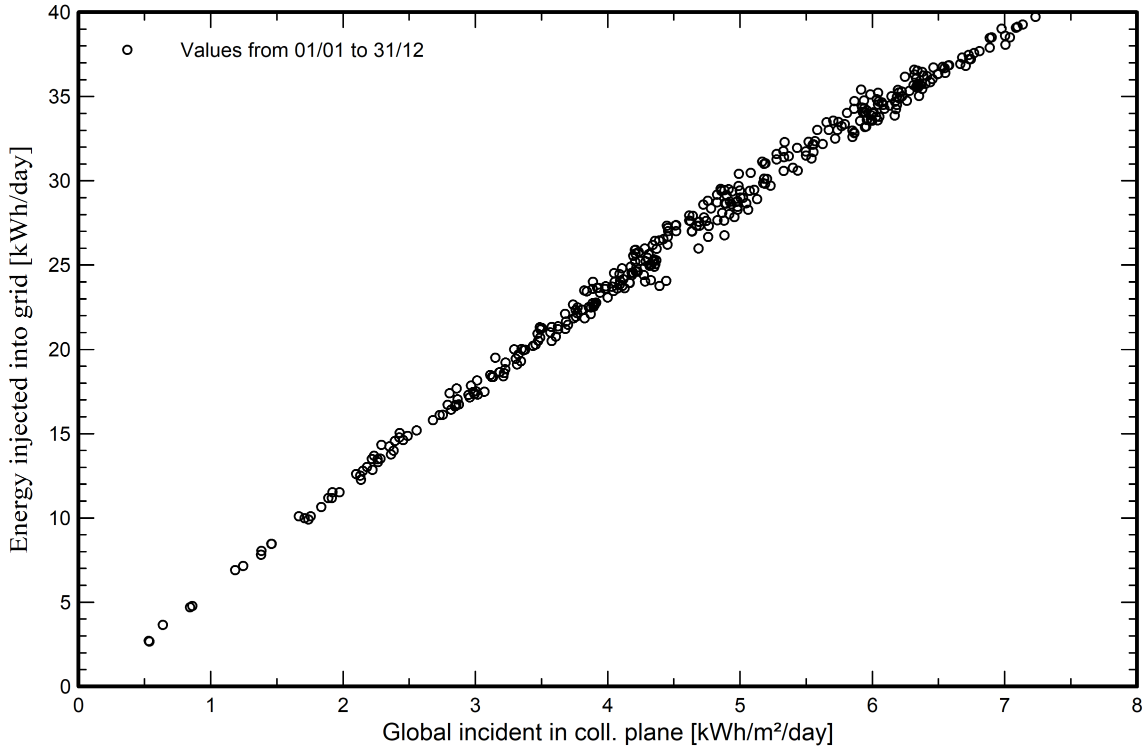
Loss diagram



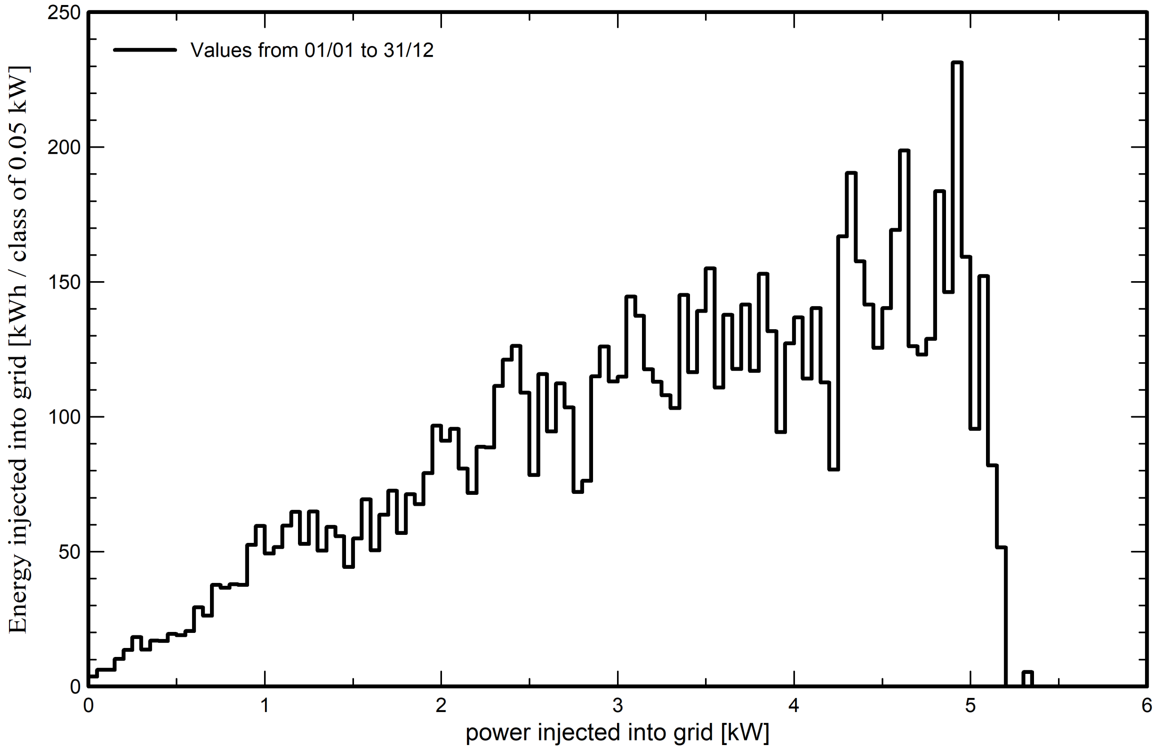
1574 kWh/m²	**Global horizontal irradiation**
+5.3%	**Global incident in coll. plane**
-2.64%	IAM factor on global
1614 kWh/m² * 44 m² coll.	**Effective irradiation on collectors**
efficiency at STC = 16.17%	PV conversion
11367 kWh	**Array nominal energy (at STC effic.)**
-0.67%	PV loss due to irradiance level
-10.21%	PV loss due to temperature
+0.75%	Module quality loss
-2.10%	Mismatch loss, modules and strings
-0.96%	Ohmic wiring loss
9903 kWh	**Array virtual energy at MPP**
-2.93%	Inverter Loss during operation (efficiency)
-0.01%	Inverter Loss over nominal inv. power
-0.19%	Inverter Loss due to max. input current
0.00%	Inverter Loss over nominal inv. voltage
-0.02%	Inverter Loss due to power threshold
0.00%	Inverter Loss due to voltage threshold
-0.09%	Night consumption
9583 kWh	**Available Energy at Inverter Output**
9583 kWh	**Energy injected into grid**

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



Single-line diagram

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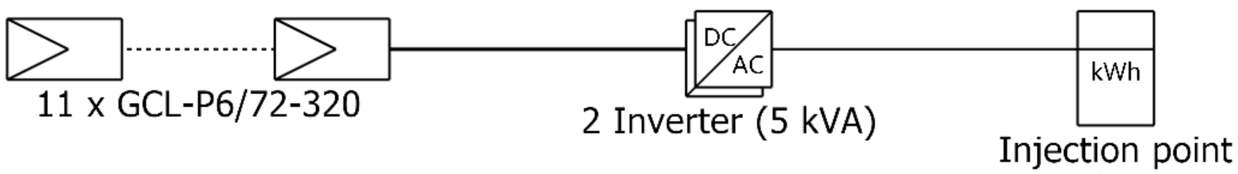

PV module	GCL-P6/72-320
Inverter	Solar Inverter RPI H2.5 Flex
String	11 x GCL-P6/72-320

kolkatawith high rise

VC0 : New simulation variant

22/10/24